Exercise of chapter 1,2,3.

Ch1Ex1:

```
public class Ch1Ex1{.
    public static void main(String[] args){.
    System.out.println("Welcome to Java. Welcome to
  Computer science. Programming is fun");
 }
}.
```

Ch1Ex2:

```
public class ch1Ex2{.
public static void main(String[] args){.
   for(int i=0; i<5; i++){
          System.out.println("Hello Java");
   }.
}}.
```

Ch1E4:

```
public class Ch1Ex4{.
public static void main(String[] args){.
System.out.println(" a            a^2     a^3 ");
System.out.println("  1           1         1 ");
System.out.println("  2           4         8 ");
System.out.println("  3           9        27 ");
System.out.println("  4          16        64 ");
}}.
```

Ch1Ex5:

```
public class Ch1Ex5{
public static void main(String[] args){
 System.out.println((9.5 * 4.5-2.5 * 3)/(45.5-3.5));}
}
```

Ch1Ex6:

```
public class Ch1Ex6{
public static void main(String[] args){
System.out.println(1+2+3+4+5+6+7+8+9);}
}
```

Ch1Ex7:

```
public class Ch1Ex7{
public static void main(String[] args){
double Pi1=4.0*(1.0-(1.0/3)+(1.0/5)-(1.0/7)+(1.0/9)-(1.0/11));
double Pi2=4.0*(1.0-(1.0/3)+(1.0/5)-(1.0/7)+(1.0/9)-(1.0/11)+(1.0/13));
System.out.println(pi1);
System.out.println(pi2);}}
```

Ch1Ex8:

```
public class Ch1Ex8{
public static void main(String[] args){
  double radius = 5.5;
  double perimeter = 2*radius*Math.PI;
  double area = radius * radius * Math.PI;
  System.out.println("Perimeter = "+ perimeter);
  System.out.println("Area = "+area);
}
}
```

Ch1Ex9:

```
public class Ch1Ex9{ .
public static void main(String[] args){ .
double width=4.5; double height = 7.9 ;
double area = width* height ;
double perimeter = 2* (width + height) ;
system.out.println("Area = "+ area) ;
System.out.println("Perimeter = "+perimeter);
}
} .
```

Ch1Ex10:

```
public class Ch1Ex10{ .
public static void main(String[] args){ .
double kilometers = 14.0 ;
double miles = kilometers/1.6 ;
double timeInHours = 45.0/60.0+30.0/3600.0 .
double averageSpeed = miles / timeInHours .
System.out.println("The average speed in miles per
hour is ");
System.out.println (averageSpeed) ;
}
}.
```

Ch1Ex11:

```
public class Ch1Ex11{
 public static void main(String[ ] args){
      double currentPopulation = 312032486;
 double secondsInYear = 365*24*60*60;
 double birthsPerYear = secondsInYear/7.0;
 double deathsPerYear = secondsInYear/13.0;
 for(int i=1; i<=5; i++){
 currentPopulation += birthsPerYear+immigrantsPerYear-
              deathsPerYear;
 System.out.println("Year "+i+" population:"+(int)
 currentPopulation);}}}
```

Ch2Ex1:

```
import java.util.Scanner;
public class Ch2Ex1{
  public static void main(String[] args){
  Scanner input = new Scanner (System.in);
   System.out.print("Enter a degree in Celsius:");
   double celsius = input.nextDouble();
   double fahrenheit = (9.0/5)* celsius+32;
 System.out.println(celsius+ "Celsius is" + fahrenheit+ "Fahrenheit");
}}
```

Ch2Ex2:

```
import java.util.Scanner;
public class Ch2Ex2{
 public static void main (String[] args){
 Scanner input = new Scanner (System.in);
 System.out.print("Enter the radius and length");
 double radius= input.nextDouble()
 double length = input.nextDouble();
```

```
double arrea = readius * readius * Math.PI ;
double volume = arrea * length;
system.out.println ("The arrea is "+arrea);
system.out.println("The volume is "+volume);}};
```

<u>Ch2Ex3:</u>

```
import java.util.scanner;
public class ch2Ex3{
public static void main(String[] args){
Scanner input = new Scanner (System.in);
System.out.print ("Enter the value for feet :");
double feet = input.nextDouble();
double meter = feet * 0.305 ;
system.out.println(feet + "feet is "+meters + "meters");}}.
```

<u>Ch2Ex9:</u>

```
import java.util.scanner;
public class ch2Ex9{
public static void main(String[] args){
scanner input = new Scanner (system.in);
system.out.print ("Enter v0, v1, and t: ");
double vo = input.nextDouble();
double v1 = input.nextDouble();
double t = input.nextDouble();

double a = (v1-vo)/t ;
System.out.println("The average accleration is"+a);

}
}.
```

Ch2Ex12:

```
import java.util.scanner;
public class ch2Ex12{ .
public static void main(String[] args){ .
Scanner input = new Scanner(System.in);
System.out.print("Enter speed and acceleration: ");
double v = input.nextDouble();
double a = input.next Double();
double length = (v*v)/(2*a);
System.out.println("The minimum runway length for this
airplane is " +length); }}.
```

Ch2Ex16:

```
import java.util.scanner;
public class ch2Ex16{.
public static void main(String[]args){ .
Scanner input = new Scanner(System.in);
system.out.print ("Enter the length of the side: ");
double s= input.nextDouble();
double area = (3* Math.pow(3,0.5)/2) * Math.pow (s,2);
System.out.println("The area of the hexagon is "+ area);
}};
```

Ch2Ex20:

```
import java.util.scanner;
public class ch2Ex20{ .
public static void main(string[].args){ .
Scanner input = new scanner (system.in);
System.out.print("Enter balance and interest rate ; ");
double balance = input.nextDouble();
double annualInterestRate
```

```
double intrest = balance * (annualInterestRate/1200);
System.out.println("The interest is"+ interest);}}.
```

Ch2Ex23:

```
import java.util.Scanner;
public class Ch2Ex23{.
 public static void main(String[] args){.
  Scanner input = new Scanner(System.in);
  System.out.print("Enter the draiving distance: ");
  double distance = input.nextDouble();
  System.out.print(" Enter miles per gallon:");
  double mpg= input.nextDouble();
  System.out.print ("Enter price per gallon: ");
  double price = input.nextDoubel();
  double cost = (distance/msg)* price;
  System.out.println ("The cost of draiving is $"+cost);}}.
```

Ch3Ex1:

```
import java.util.Scanner;
public class Ch3Ex1{
 public static void main(String[] args){.
  Scanner input = new Scanner(System.in);
  System.out.praint("Enter a,b,c: ");
  double a= input.nextDouble();
  double b= input.nextDouble();
  double c = input.nextDouble();
  double discriminant = Math.pow(b,2) - 4* a*c;
```

```
if(discriminant >0){
double r1 = (-b+Math.pow(discriminant, 0.05)) / (2* a);
double r2 = (-b-Math.pow(discriminant,0.05))/(2* a);
System.out.println("r1 and r2 "+ r1 + "and" + r2); }
else if (discriminant ==0){
  double r1 = -b/(2*a);
  System.out.println("Root is "+r1);}
else{
 System.out.println("No real roots"); }}}
```

Ch3Ex3:

```
import java.util.Scanner;
public class Ch3Ex3{
 public static void main (String[] args){
  Scanner input =new Scanner (system.in);
 System.out.print("Enter a, b, c, d, e, f: ");
double a= input.nextDouble();
double b = input.nextDouble();
double c = input.nextDouble();
double d= input.nextDouble();
double e = input.nextDouble();
double f= input.nextDouble();
double det = a*d - b*c;
if (det ==0)
  System.out.println("The equation has no solution").
else
  double x = (e*d - b*f)/det;
  double y = (a*f - e*c)/det;
  System.out.println("x is "+x+"and y is"+y);}}}
```

Ch3Ex8:

```
import java.util.Scanner;
public class Ch3Ex8{
public static void main(String[] args){
Scanner input = new Scanner(System.in);
System.out.print("Enter three integers: ");

int n1 = input.nextInt();
 int n2 = input.nextInt();
 int n3 = input.nextInt();

 int temp ;
 if(n1>n2){temp= n1; n1=n2; n2 = temp;}
 if (n2>n3){temp= n2; n2= n3; n3 =temp;}
 if (n1
 System.out.println(n1+" "+ n2+" "+n3);}}
```

Ch3Ex10:

```
import java.util.Scanner;
public class Ch3Ex10{
  public static void main(String[] args){
   Scanner input =new Scanner(System.in);
   int n1=(int)(Math.random()* 100);
   int n2 = (int) (Math.random()* 100);
  System.out.print("What is "+n1 "+"+n2 + "?");
  int   answer= input.nextInt();
  if (n1+n2 == answer)
      System.out.println("You are correct!");
  else
      System.out.println(" Wrong Answer ");}}
```

Ch3Ex14:

```
import java.util.scanner;
public class Ch3Ex14{ .
public static void main(String[] args){.
 Scanner input = new Scanner(System.in);
 int coin = (int) (Math.random()* 2);
 System.out.print("Guess head(0) or tail(1): ");
int guess = input.nextInt();
if (guess == coin)
    system.out.println("Correct");
else
    system.out.println("Incorrect"); }}.
```

Ch3Ex16:

~~import java.util.~~

```
public class Ch3Ex16{ .
  public static void main(String[] args){.
 double x = Math.random()* 100-50;
 double y = Math.random() * 200-100;
 System.out.println("Random coordinate: ("+x+", "+y
        +")"); }}.
```

Ch3Ex19:

```
import java.util.scanner;
public class Ch3Ex19{ .
public static void main(string[] args){.
 Scanner input = new Scanner(System.in);
System.out.print ("Enter three edges of the triangle: ");
double a = input.nextDouble();
```

```
double b = input(next Double();
double c = input.nextDouble();
if(a+b>c && a+c>b && b+c>a){
  system.out.println ("Perimeter is "+ (a+b+c));}
else{
 system.out.println ("Input is invalid");
}}}
```

Ch3Ex22:

```
import java.util.scanner;
public class Ch3Ex22 {
public static void main(string[] args){
 scanner input = new Scanner(System.in);
System.out.print("Enter a point with two coordinates: ");
double x = input.nextDouble();
double y = input.nextDouble();
double distance = Math.pow (x*x +y*y ,0.5);
if (distance <= 10){
  System.out.println("Point ("+x+", "+y+") is the circle");
 else
  system.out.println("Point("x+", "+y+") is not circle");}}
```

Ch3Ex26:

```
import java.util.scanner;
public class ch3Ex26{
public static void main (String[] args){
 Scanner input = new Scanner(System.in);
System.out.print("Enter an integer: ");
```

```
System.out.print("Enter an integer: ");
int num = input.nextInt();
System.out.println("Is "+num+ "divisible by 5 and 6?"+
    (num%5==5 && num%6==0));
System.out.println("Is "+num+ "divisible by 5 or 6?"+
 (num%5==0|| num%6==0));
System.out.println("Is "+num+ "divisible by 5 or
6, but not both?"+(num%5==0 ^ num%6==0));}}
```

Ch3Ex31.java:

```
import java.util.Scanner;
public class Ch3Ex31{
public static void main(String[] args){
 Scanner input = new Scanner(System.in);
 System.out.print("Enter the exchange rate from dollar toRMB
double rate= input.nextDouble();
System.out.print("Enter 0 to convert to RMB and 1 vice versa:")
 int direction= input.nextInt();
 if (direction==0){
   System.out.print("Enter the dollar amount: ");
   double dollars = input.nextDouble();
   double rmb= dollars * rate;
   System.out.println("$"+dollars+"is "+rmb +" yuan");
}else if(direction==1){
 System.out.
```